

- Definitions of sets, characteristics of sets, and how they enter into definitions of probability.
- Introduction of probability defined through sets and through the use of the relative frequency concept.
- Development of special kinds of probability, such as applied to events, multiple events (joint probability), and events dependent on each other (conditional probability).
- Introduction of the statistical independence of events.
- Discussions of how to combine several separate random experiments such that they may be taken as a single (combined) experiment. Through Bernoulli trials these topics were applied to various practical problems involving success and failure outcomes of an experiment.

This chapter's topics form a firm basis to proceed to the important concept of a random variable in the next chapter.

PROBLEMS

1.1-1. Specify the following sets by the rule method.

$$A = \{1, 2, 3\}, B = \{8, 10, 12, 14\}, C = \{1, 3, 5, 7, \dots\}$$

1.1-2. Use the tabular method to specify a class of sets for the sets of Problem 1.1-1.

1.1-3. State whether the following sets are countable or uncountable, or finite or infinite. $A = \{1\}$, $B = \{x = 1\}$, $C = \{0 < \text{integers}\}$, $D = \{\text{children in public school No. 5}\}$, $E = \{\text{girls in public school No. 5}\}$, $F = \{\text{girls in class in public school No. 5 at 3:00 A.M.}\}$, $G = \{\text{all lengths not exceeding one meter}\}$, $H = \{-25 \leq x \leq -3\}$, $I = \{-2, -1, 1 \leq x \leq 2\}$.

1.1-4. For each set of Problem 1.1-3, determine if it is equal to, or a subset of, any of the other sets.

1.1-5. State every possible subset of the set of letters $\{a, b, c, d\}$.

1.1-6. A thermometer measures temperatures from -40 to 130°F (-40 to 54.4°C).
 (a) State a universal set to describe temperature measurements. Specify subsets for:
 (b) Temperature measurements not exceeding water's freezing point, and
 (c) Measurements exceeding the freezing point but not exceeding 100°F (37.8°C).

*1.1-7. Prove that a set with N elements has 2^N subsets.

1.1-8. A random noise voltage at a given time may have any value from -10 to 10 V .
 (a) What is the universal set describing noise voltage?
 (b) Find a set to describe the voltages available from a half-wave rectifier for positive voltages that has a linear output-input voltage characteristic.
 (c) Repeat parts (a) and (b) if a dc voltage of -3 V is added to the random noise.

- 1.1-9.** Use the tabular method to define a set A that contains all integers with magnitudes not exceeding 7. Define a second set B having odd integers larger than -2 and not larger than 5. Determine if $A \subset B$ and $B \subset A$.
- 1.1-10.** A set A has three elements a_1, a_2 , and a_3 . Determine all possible subsets of A .
- 1.1-11.** Specify, by both the tabular and rule methods, each of the following sets: (a) all integers between 1 and 9, (b) all integers from 1 to 9, (c) the five values of equivalent resistance for n identical $10\text{-}\Omega$ resistors in parallel where $n = 1, 2, \dots, 5$, and (d) the six values of equivalent resistance for n identical $2.2\text{-}\Omega$ resistors in series where $n = 1, 2, \dots, 6$.
- 1.1-12.** A box contains 100 capacitors (universal set) of which 40 are $0.01\text{ }\mu\text{F}$ with a 100-V voltage rating, 35 are $0.1\text{ }\mu\text{F}$ at a rating of 50 V , and 25 are $1.0\text{ }\mu\text{F}$ and have a 10-V rating. Determine the number of elements in the following sets:
 (a) $A = \{\text{capacitors with capacitance} \geq 0.1\text{ }\mu\text{F}\}$
 (b) $B = \{\text{capacitors with voltage rating} > 5\text{ V}\}$
 (c) $C = \{\text{capacitors with both capacitance} \geq 0.1\text{ }\mu\text{F and voltage rating} \geq 50\text{ V}\}$.
- 1.2-1.** Show that $C \subset A$ if $C \subset B$ and $B \subset A$.
- 1.2-2.** Two sets are given by $A = \{-6, -4, -0.5, 0, 1.6, 8\}$ and $B = \{-0.5, 0, 1, 2, 4\}$. Find:
 (a) $A - B$ (b) $B - A$ (c) $A \cup B$ (d) $A \cap B$
- 1.2-3.** A universal set is given as $S = \{2, 4, 6, 8, 10, 12\}$. Define two subsets as $A = \{2, 4, 10\}$ and $B = \{4, 6, 8, 10\}$. Determine the following:
 (a) $\bar{A} = S - A$ (b) $A - B$ and $B - A$ (c) $A \cup B$ (d) $A \cap B$
 (e) $\bar{A} \cap B$
- 1.2-4.** Using Venn diagrams for three sets A, B, C , shade the areas corresponding to the sets:
 (a) $(A \cup B) - C$ (b) $\bar{B} \cap A$ (c) $A \cap B \cap C$ (d) $(\overline{A \cup B}) \cap C$
- 1.2-5.** Sketch a Venn diagram for three events where $A \cap B \neq \emptyset$, $B \cap C \neq \emptyset$, $C \cap A \neq \emptyset$, but $A \cap B \cap C = \emptyset$.
- 1.2-6.** Use Venn diagrams to show that the following identities are true:
 (a) $(\overline{A \cup B}) \cap C = C - [(A \cap C) \cup (B \cap C)]$
 (b) $(A \cup B \cup C) - (A \cap B \cap C) = (\bar{A} \cap B) \cup (\bar{B} \cap C) \cup (\bar{C} \cap A)$
 (c) $(\overline{A \cap B \cap C}) = \bar{A} \cup \bar{B} \cup \bar{C}$
- 1.2-7.** Use Venn diagrams to prove De Morgan's laws $(\overline{A \cup B}) = \bar{A} \cap \bar{B}$ and $(\overline{A \cap B}) = \bar{A} \cup \bar{B}$.
- 1.2-8.** A universal set is $S = \{-20 < s \leq -4\}$. If $A = \{-10 \leq s \leq -5\}$ and $B = \{-7 < s < -4\}$, find:
 (a) $A \cup B$
 (b) $A \cap B$
 (c) A third set C such that the sets $A \cap C$ and $B \cap C$ are as large as possible while the smallest element in C is -9 .

(d) What is the set $A \cap B \cap C$?

1.2-9. Use De Morgan's laws to show that:

(a) $\overline{A \cap (B \cup C)} = (\bar{A} \cup \bar{B}) \cap (\bar{A} \cup \bar{C})$

(b) $\overline{(A \cap B \cap C)} = \bar{A} \cup \bar{B} \cup \bar{C}$

In each case check your results using a Venn diagram.

1.2-10. Shade Venn diagrams to illustrate each of the following sets:

(a) $(A \cup \bar{B}) \cap \bar{C}$, (b) $(\bar{A} \cap \bar{B}) \cup \bar{C}$, (c) $(A \cup \bar{B}) \cup (C \cap D)$,

(d) $(A \cap B \cap \bar{C}) \cup (\bar{B} \cap C \cap D)$.

1.2-11. A universal set S is composed of all points in a rectangular area defined by $0 \leq x \leq 3$ and $0 \leq y \leq 4$. Define three sets by $A = \{y \leq 3(x-1)/2\}$, $B = \{y \geq 1\}$, and $C = \{y \geq 3-x\}$. Shade in Venn diagrams corresponding to the sets (a) $A \cap B \cap C$, and (b) $C \cap B \cap \bar{A}$.

1.2-12. The take-off roll distance for aircraft at a certain airport can be any number from 80 m to 1750 m. Propeller aircraft require from 80 m to 1050 m while jets use from 950 m to 1750 m. The overall runway is 2000 m.

(a) Determine sets A , B , and C defined as "propeller aircraft take-off distances," "jet aircraft take-off distances," and "runway length safety margin," respectively.

(b) Determine the set $A \cap B$ and give its physical significance.

(c) What is the meaning of the set $\overline{A \cup B}$?

(d) What are the meanings of the sets $\overline{A \cup B \cup C}$ and $A \cup B$?

1.2-13. Prove that De Morgan's law (1.2-13) can be extended to N events A_i , $i = 1, 2, \dots, N$ as follows:

$$\overline{(A_1 \cap A_2 \cap \dots \cap A_N)} = (\bar{A}_1 \cup \bar{A}_2 \cup \dots \cup \bar{A}_N)$$

1.2-14. Work Problem 1.2-13 for (1.2-12) to prove

$$\overline{(A_1 \cup A_2 \cup \dots \cup A_N)} = (\bar{A}_1 \cap \bar{A}_2 \cap \dots \cap \bar{A}_N)$$

1.2-15. Sets $A = \{1 \leq s \leq 14\}$, $B = \{3, 6, 14\}$, and $C = \{2 < s \leq 9\}$ are defined on a sample space S . State if each of the following conditions is true or false.

(a) $C \subset B$, (b) $C \subset A$, (c) $B \cap C = \emptyset$, (d) $C \cup B = S$,
(e) $\bar{S} = \emptyset$, (f) $A \cap \bar{S} = \emptyset$, and (g) $C \subset A \subset B$.

1.2-16. Draw Venn diagrams and shade the areas corresponding to the sets
(a) $(A \cup B \cup C) \cap (\bar{A} \cup \bar{B} \cup \bar{C})$, and (b) $[(A \cup \bar{B}) \cap C] \cup (\bar{A} \cup \bar{C})$.

1.2-17. Work Problem 1.2-16 except assume sets (a) $(A \cap B \cap C) \cup (\bar{A} \cup \bar{B} \cup \bar{C})$,
(b) $B - (A \cap B)$, and (c) $(A \cap B) \cup (A \cap C) \cup (B \cap C) - (A \cap B \cap C)$.

1.3-1. A die is tossed. Find the probabilities of the events $A = \{\text{odd number shows up}\}$, $B = \{\text{number larger than 3 shows up}\}$, $A \cup B$, and $A \cap B$.

1.3-2. In a game of dice, a "shooter" can win outright if the sum of the two numbers showing up is either 7 or 11 when two dice are thrown. What is his probability of winning outright?

- 1.3-3. A pointer is spun on a fair wheel of chance having its periphery labeled from 0 to 100.
- What is the sample space for this experiment?
 - What is the probability that the pointer will stop between 20 and 35?
 - What is the probability that the wheel will stop on 58?
- 1.3-4. An experiment has a sample space with 10 equally likely elements $S = \{a_1, a_2, \dots, a_{10}\}$. Three events are defined as $A = \{a_1, a_5, a_9\}$, $B = \{a_1, a_2, a_6, a_9\}$, and $C = \{a_6, a_9\}$. Find the probabilities of:
- $A \cup C$
 - $B \cup \bar{C}$
 - $A \cap (B \cup C)$
 - $\bar{A} \cup \bar{B}$
 - $(A \cup B) \cap C$
- 1.3-5. Let A be an arbitrary event. Show that $P(\bar{A}) = 1 - P(A)$.
- 1.3-6. An experiment consists of rolling a single die. Two events are defined as: $A = \{\text{a 6 shows up}\}$ and $B = \{\text{a 2 or a 5 shows up}\}$.
- Find $P(A)$ and $P(B)$.
 - Define a third event C so that $P(C) = 1 - P(A) - P(B)$.
- 1.3-7. In a box there are 500 colored balls: 75 black, 150 green, 175 red, 70 white, and 30 blue. What are the probabilities of selecting a ball of each color?
- 1.3-8. A single card is drawn from a 52-card deck.
- What is the probability that the card is a jack?
 - What is the probability the card will be a 5 or smaller?
 - What is the probability that the card is a red 10?
- 1.3-9. A pair of fair dice are thrown in a gambling problem. Person A wins if the sum of numbers showing up is six or less *and* one of the dice shows four. Person B wins if the sum is five or more *and* one of the dice shows a four. Find: (a) The probability that A wins, (b) the probability of B winning, and (c) the probability that both A and B win.
- 1.3-10. You (person A) and two others (B and C) each toss a fair coin in a two-step gambling game. In step 1 the person whose toss is not a match to either of the other two is "odd man out." Only the remaining two whose coins match go on to step 2 to resolve the ultimate winner.
- What is the probability you will advance to step 2 after the first toss?
 - What is the probability you will be out after the first toss?
 - What is the probability that no one will be out after the first toss?
- 1.3-11. A particular electronic device is known to contain only 10-, 22-, and 48- Ω resistors, but these resistors may have 0.25-, 0.5-, or 1-W ratings, depending on how purchases are made to minimize cost. Historically, it is found that the probabilities of the 10- Ω resistors being 0.25, 0.5, or 1 W are 0.08, 0.10, and 0.01, respectively. For the 22- Ω resistors the similar probabilities are 0.20, 0.26, and 0.05. It is also historically found that the probabilities are 0.40, 0.51, and 0.09 that any resistors are 0.25, 0.50, and 1 W, respectively. What are the probabilities that the 48- Ω resistors are (a) 0.25, (b) 0.50, and (c) 1 W?

- 1.3-12.** For the sample space defined in Example 1.3-2 find the probabilities that: (a) one die will show a 2 and the other will show a 3 or larger, and (b) the sum of the two numbers showing up will be 4 or less or will be 10 or more.
- 1.3-13.** In a game two dice are thrown. Let one die be "weighted" so that a 4 shows up with probability $\frac{2}{7}$, while its other numbers all have probabilities of $\frac{1}{7}$. The same probabilities apply to the other die except the number 3 is "weighted." Determine the probability the shooter will win outright by having the sum of the numbers showing up be 7. What would be the probability for fair dice?
- 1.4-1.** Two cards are drawn from a 52-card deck (the first is not replaced).
 (a) Given the first card is a queen, what is the probability that the second is also a queen?
 (b) Repeat part (a) for the first card a queen and the second card a 7.
 (c) What is the probability that both cards will be a queen?
- 1.4-2.** An ordinary 52-card deck is thoroughly shuffled. You are dealt four cards up. What is the probability that all four cards are sevens?
- 1.4-3.** For the resistor selection experiment of Example 1.4-1, define event D as "draw a 22- Ω resistor," and E as "draw a resistor with 10% tolerance." find $P(D)$, $P(E)$, $P(D \cap E)$, $P(D|E)$, and $P(E|D)$.
- 1.4-4.** For the resistor selection experiment of Example 1.4-1, define two mutually exclusive events B_1 and B_2 such that $B_1 \cup B_2 = S$.
 (a) Use the total probability theorem to find the probability of the event "select a 22- Ω resistor," denoted D .
 (b) Use Bayes' theorem to find the probability that the resistor selected had 5% tolerance, given it was 22 Ω .
- 1.4-5.** In three boxes there are capacitors as shown in Table P1.4-5. An experiment consists of first randomly selecting a box, assuming each has the same likelihood of selection, and then selecting a capacitor from the chosen box.
 (a) What is the probability of selecting a 0.01- μF capacitor, given that box 2 is selected?
 (b) If a 0.01- μF capacitor is selected, what is the probability it came from box 3? (*Hint:* Use Bayes' and total probability theorems.)

TABLE P1.4-5
Capacitors

Value (μF)	Number in box			Totals
	1	2	3	
0.01	20	95	25	140
0.1	55	35	75	165
1.0	70	80	145	295
Totals	145	210	245	600

- 1.4-6.** For Problem 1.4-5, list the nine conditional probabilities of capacitor selection, given certain box selections.
- 1.4-7.** Rework Example 1.4-2 if $P(B_1) = 0.6$, $P(B_2) = 0.4$, $P(A_1|B_1) = P(A_2|B_2) = 0.95$, and $P(A_2|B_1) = P(A_1|B_2) = 0.05$.
- 1.4-8.** Rework Example 1.4-2 if $P(B_1) = 0.7$, $P(B_2) = 0.3$, $P(A_1|B_1) = P(A_2|B_2) = 1$ and $P(A_2|B_1) = P(A_1|B_2) = 0$. What type of channel does this system have?
- 1.4-9.** A company sells high fidelity amplifiers capable of generating 10, 25, and 50 W of audio power. It has on hand 100 of the 10-W units, of which 15% are defective, 70 of the 25-W units with 10% defective, and 30 of the 50-W units with 10% defective.
- What is the probability that an amplifier sold from the 10-W units is defective?
 - If each wattage amplifier sells with equal likelihood, what is the probability of a randomly selected unit being 50 W and defective?
 - What is the probability that a unit randomly selected for sale is defective?
- 1.4-10.** A missile can be accidentally launched if two relays A and B both have failed. The probabilities of A and B failing are known to be 0.01 and 0.03, respectively. It is also known that B is more likely to fail (probability 0.06) if A has failed.
- What is the probability of an accidental missile launch?
 - What is the probability that A will fail if B has failed?
 - Are the events “ A fails” and “ B fails” statistically independent?
- *1.4-11.** The communication system of Example 1.4-2 is to be extended to the case of three transmitted symbols 0, 1, and 2. Define appropriate events A_i and B_i , $i = 1, 2, 3$, to represent symbols after and before the channel, respectively. Assume channel transition probabilities are all equal at $P(A_i|B_j) = 0.1$, $i \neq j$, and are $P(A_i|B_j) = 0.8$ for $i = j = 1, 2, 3$, while symbol transmission probabilities are $P(B_1) = 0.5$, $P(B_2) = 0.3$, and $P(B_3) = 0.2$.
- Sketch the diagram analogous to Fig. 1.4-2.
 - Compute received symbol probabilities $P(A_1)$, $P(A_2)$, and $P(A_3)$.
 - Compute the a posteriori probabilities for this system.
 - Repeat parts (b) and (c) for all transmission symbol probabilities equal. Note the effect.
- 1.4-12.** A pharmaceutical product consists of 100 pills in a bottle. Two production lines used to produce the product are selected with probabilities 0.45 (line one) and 0.55 (line two). Each line can overfill or underfill bottles by at most 2 pills. Given that line one is observed, the probabilities are 0.02, 0.06, 0.88, 0.03, and 0.01 that the numbers of pills in a bottle will be 102, 101, 100, 99, and 98, respectively. For line two, the similar respective probabilities are 0.03, 0.08, 0.83, 0.04, and 0.02.
- Find the probability that a bottle of the product will contain 102 pills. Repeat for 101, 100, 99, and 98 pills.
 - Given that a bottle contains the correct number of pills, what is the probability it came from line one?
 - What is the probability that a purchaser of the product will receive less than 100 pills?

- 1.4-13.** A manufacturing plant makes radios that each contains an integrated circuit (IC) supplied by three sources A , B , and C . The probability that the IC in a radio came from one of the sources is $\frac{1}{3}$, the same for all sources. ICs are known to be defective with probabilities 0.001, 0.003, and 0.002 for sources A , B , and C , respectively.
- What is the probability any given radio will contain a defective IC?
 - If a radio contains a defective IC, find the probability it came from source A . Repeat for sources B and C .
- 1.4-14.** There are three special decks of cards. The first, deck D_1 , has all 52 cards of a regular deck. The second, D_2 , has only the 16 face cards of a regular deck (only 4 each of jacks, queens, kings, and aces). The third, D_3 , has only the 36 numbered cards of a regular deck (4 twos through 4 tens). A random experiment consists of first randomly choosing one of the three decks, then second, randomly choosing a card from the chosen deck. If $P(D_1) = \frac{1}{2}$, $P(D_2) = \frac{1}{3}$, and $P(D_3) = \frac{1}{6}$, find the probabilities: (a) of drawing an ace, (b) of drawing a three, and (c) of drawing a red card.
- 1.5-1.** Determine whether the three events A , B , and C of Example 1.4-1 are statistically independent.
- 1.5-2.** List the various equations that four events A_1 , A_2 , A_3 , and A_4 must satisfy if they are to be statistically independent.
- *1.5-3.** Given that two events A_1 and A_2 are statistically independent, show that:
- $\bar{A}_1$ is independent of $\bar{A}_2$
 - $\bar{A}_1$ is independent of A_2
 - A_1 is independent of $\bar{A}_2$
- 1.5-4.** Show that there are $2^N - N - 1$ equations required in (1.5-6). (*Hint:* Recall that the binomial coefficient is the number of combinations of N things taken n at a time.)
- 1.5-5.** In a communication system the signal sent from point a to point b arrives by two paths in parallel. Over each path the signal passes through two repeaters (in series). Each repeater in one path has a probability of failing (becoming an open circuit) of 0.005. This probability is 0.008 for each repeater on the other path. All repeaters fail independently of each other. Find the probability that the signal will not arrive at point b .
- 1.5-6.** Work Problem 1.5-5, except assume the paths and repeaters of Figure P1.5-6, where the probabilities of the repeaters' failing (independently) are $p_1 = P(R_1) = 0.005$, $p_2 = P(R_2) = P(R_3) = P(R_4) = 0.01$, and $p_3 = P(R_5) = P(R_6) = 0.05$.
- 1.6-1.** An experiment consists of randomly selecting one of five cities on Florida's west coast for a vacation. Another experiment consists of selecting at random one of four acceptable motels in which to stay. Define sample spaces S_1 and S_2 for the two experiments and a combined space $S = S_1 \times S_2$ for the combined experiment having the two subexperiments.

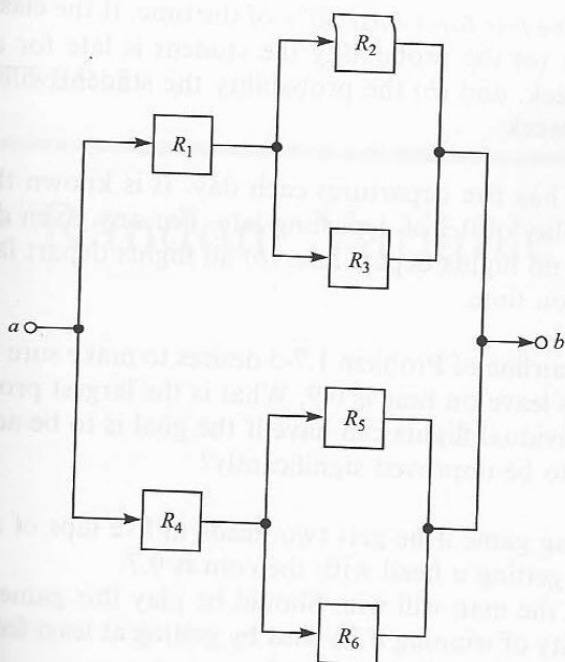


FIGURE P1.5-6

- 1.6-2.** Sketch the area in the combined sample space of Example 1.6-3 corresponding to the event $A \times B$ where:
- $A = \{10 < x \leq 15\}$ and $B = \{20 < y \leq 50\}$
 - $A = \{x = 40\}$ and $B = \{5 < y \leq 40\}$
- 1.6-3.** The six sides of a fair die are numbered from 1 to 6. The die is rolled four times. How many sequences of the four resulting numbers are possible?
- 1.6-4.** In a 5-card poker game, a player is dealt 5 cards. How many poker hands are possible for an ordinary 52-card deck?
- 1.7-1.** A production line manufactures 5-gal (18.93-liter) gasoline cans to a volume tolerance of 5%. The probability of any one can being out of tolerance is 0.03. If four cans are selected at random:
- What is the probability they are all out of tolerance?
 - What is the probability of exactly two being out?
 - What is the probability that all are in tolerance?
- 1.7-2.** Spacecraft are expected to land in a prescribed recovery zone 80% of the time. Over a period of time, six spacecraft land.
- Find the probability that none lands in the prescribed zone.
 - Find the probability that at least one will land in the prescribed zone.
 - The landing program is called successful if the probability is 0.9 or more that three or more out of six spacecraft will land in the prescribed zone. Is the program successful?
- 1.7-3.** In the submarine problem of Example 1.7-1, find the probabilities of sinking the carrier when fewer ($N = 2$) or more ($N = 4$) torpedoes are fired.

- 1.7-4.** A student is known to arrive late for a class 40% of the time. If the class meets five times each week find: (a) the probability the student is late for at least three classes in a given week, and (b) the probability the student will not be late at all during a given week.
- 1.7-5.** An airline in a small city has five departures each day. It is known that any given flight has a probability of 0.3 of departing late. For any given day find the probabilities that: (a) no flights depart late, (b) all flights depart late, and (c) three or more depart on time.
- 1.7-6.** The local manager of the airline of Problem 1.7-5 desires to make sure that the probability that all flights leave on time is 0.9. What is the largest probability of being late that the individual flights can have if the goal is to be achieved? Will the operation have to be improved significantly?
- 1.7-7.** A man wins in a gambling game if he gets two heads in five flips of a biased coin. The probability of getting a head with the coin is 0.7.
(a) Find the probability the man will win. Should he play this game?
(b) What is his probability of winning if he wins by getting at least four heads in five flips? Should he play this new game?
- *1.7-8.** A rifleman can achieve a "marksman" award if he passes a test. He is allowed to fire six shots at a target's bull's eye. If he hits the bull's eye with at least five of his six shots he wins a set. He becomes a marksman only if he can repeat the feat three times straight, that is, if he can win three straight sets. If his probability is 0.8 of hitting a bull's eye on any one shot, find the probabilities of his: (a) winning a set, and (b) becoming a marksman.
- 1.7-9.** A ship can successfully arrive at its destination if its engine and its satellite navigation system do not fail en route. If the engine and navigation system are known to fail independently with respective probabilities of 0.05 and 0.001, what is the probability of a successful arrival?
- 1.7-10.** At a certain military installation six similar radars are placed in operation. It is known that a radar's probability of failing to operate before 500 hours of "on" time have accumulated is 0.06. What are the probabilities that before 500 hours have elapsed, (a) all will operate, (b) all will fail, and (c) only one will fail?
- 1.7-11.** A particular model of automobile is recalled to fix a mechanical problem. The probability that a car will be properly repaired is 0.9. During the week a dealer has eight cars to repair.
(a) What is the probability that two or more of the eight cars will have to be repaired more than once?
(b) What is the probability all eight cars will be properly repaired?
- 1.7-12.** In a large hotel it is known that 99% of all guests return room keys when checking out. If 250 engineers check out after a large conference, what is the probability that not more than three will fail to return their keys? [*Hint*: Use the approximation of (1.7-8).]